



Mathieu ASTRUC

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EXPERIENCE

- Airbus** Mar. 2026 – Sep. 2026
Applied AI Engineer - Master Thesis *France*
- **Built every AI feature** of Airbus's export-control new license reference tool, end-to-end, in a safety-critical environment. Built a hybrid RAG over **10k+ export licenses** and regulatory docs (Qwen3 embeddings, FAISS), routing between a LangChain path and a NL-to-SQL path to guarantee **zero generative approximation**.
 - Engineered the document-ingestion pipeline from scratch, pairing vision-language models (**Gemma 4 31B + olmOCR 2 7B**) to parse complex regulatory documents into structured, machine-readable data.
 - Shipped to production, now serving **80+ daily users**; presented live to **100+ employees** and delivered AI POCs with cross-functional teams, driving enterprise AI adoption.
- NTNU – Norwegian University of Science and Technology** May 2025 – Aug. 2025
Research Engineer - Intern *Norway*
- Developed an embedded AI interaction stack for a humanoid robot, combining real-time gesture recognition, computer vision and a fine-tuned LLM for domain-specific dialogue.
 - Optimized on-device inference latency through GPU pipeline tuning, enabling real-time responses under tight hardware and runtime constraints.
 - Built a Human-in-the-Loop framework for edge cases and robustness; the gesture-recognition work was accepted as a **first-author paper at HCI International 2026** (Springer).
- Banque de France** May 2024 – Aug. 2024
Data Scientist - Intern *France*
- Automated scraping workflows aggregating unstructured public/financial data from multiple sources.
 - Combined OCR, Speech-to-Text and LLMs to digitize raw audio/visual content, feeding a downstream RAG system used by internal data scientist and analyst.
- AI Content Creator & Community Founder** 2025 – Present
Founder, Editor & Educator – French-speaking AI community (Discord, YouTube, Instagram, TikTok, Facebook)
- Founded and grew a French-speaking AI community to **40k+ subscribers** from scratch, making applied ML, LLMs and AI agents accessible through hands-on technical content.
 - Produced and edited technical AI videos end-to-end, generating nearly **10M views** across platforms owning the full production chain from scripting to motion design and high-retention editing.

SELECTED PROJECTS

- Mistral Vibe Hackathon – #1 Track Winner & 3rd Place Overall – Paris** 2026
- Built **Impostral** in 8 hours, a real-time social deduction game where autonomous Mistral agents infiltrate a group of humans and attempt to survive by acting human.
 - Designed a multiplayer social Turing test in which humans and AI agents answer personal questions before voting to eliminate the suspected AI. [Watch the demo.](#)
- Publication – HCI International 2026, Montreal (Springer Conference Proceedings)** 2025 – 2026
- Lead author of an accepted HRI paper on an optimized real-time computer vision architecture for gesture recognition, integrating MediaPipe landmarks, lightweight ML classifiers, and low-latency robot actuation.

EDUCATION

- Esaip – Master of Engineering, Data Science (Grade A - Top 10%)** Sep. 2021 – Sep. 2026
Engineer's Degree (5-year program) - Optimization algorithms, Multicriteria Optimization, ML, Image Processing, Reinforcement Learning, Explainable AI, Probabilistic Modeling... *France*
- Universidad Politécnica de Madrid** Sep. 2025 – Feb. 2026
Study Abroad – Machine Learning, cloud computing, large-scale data architectures *Spain*
- SeAMK – University of Applied Sciences** Jan. 2024 – Jun. 2024
Study Abroad – Machine Learning, Deep Learning, Software Development, Embedded Systems *Finland*

SKILLS

- Programming Languages** Python, C, C++, SQL, Bash
- Data Science & Machine Learning** PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, Transformers
- LLM engineering** LangChain, LlamaIndex, Agent workflows, Fine-tuning (LoRA / QLoRA), Vector DBs (FAISS, Chroma)
- Specializations** NLP, RAG, Computer Vision, Multimodal AI, Human Robot Interaction
- Inference & Deployment** vLLM, llama.cpp, Hugging Face TGI, FastAPI, Docker, Kubernetes
- Content & Media** Video editing & motion design (Premiere Pro / AI editing agents), technical storytelling, community building
- Tools & Frameworks** Python (FastAPI), JavaScript (Next.js), TypeScript (React)